Week 12+ (Optional) — Math Kangaroo-Style Stretch — Answer Key

Total: 60 points (5 × 3 + 5 × 4 + 5 × 5).

Tier 1 — 3 points each

Q1. B) △ — The pattern alternates ○, △, ○, △, ○, so the next shape is △.

Q2. B) 5 — The big triangle is divided into 4 small triangles (3 "corner" triangles plus 1 "upside-down" triangle in the middle). Counting those 4, plus the big triangle itself as a 5th, gives 5 triangles in total. No other combination of the pieces forms a triangle.

Q3. C) 4 — Worst case, you could pull out all 3 red balls first. The 4th ball you take is then guaranteed to be blue (since there are only 2 colours and you've exhausted the larger colour group). In general: the answer is (the bigger group's count) + 1 = 3 + 1 = 4.

Q4. C — Option C correctly mirrors the figure: the pole stays right next to the mirror line, the flag flips to point the opposite way, and the dot stays at the same height on the other side. (A just copies the shape without flipping it — the flag still points the same way, which a mirror would never do. B flips it top-to-bottom instead of left-to-right. D spins it 180°, which flips it both ways at once.)

Q5. B) 2nd — Spike = 7th. Jerry finished 3 places ahead of Spike: 7 − 3 = 4th. Tom finished 2 places ahead of Jerry: 4 − 2 = 2nd.

Tier 2 — 4 points each

Q6. B) 5 — The grid has 4 small unit squares, plus the big 2×2 square itself, giving 5 squares in total.

Q7. B) 10 — Let the smaller number be x; the larger is $4x$. Then $x + 4x = 50 \rightarrow 5x = 50 \rightarrow x = 10$.

Q8. C) 75 g — 1 orange = 2 kiwis = 2 × 50 g = 100 g. Since apple + banana = orange + kiwi = 100 g + 50 g = 150 g, and apple = banana, each one weighs 150 ÷ 2 = 75 g.

Q9. B) 6 — To get from the dot to the star you need exactly 2 "right" moves and 2 "down" moves, in any order. The number of distinct orders is "4 choose 2" = 6: RRDD, RDRD, RDDR, DRRD, DRDR, DDRR.

Q10. C) 11 — Working backwards: 45 ÷ 3 = 15 (this was "2× the number, minus 7"); 15 + 7 = 22 (this was "2× the number"); 22 ÷ 2 = 11.

Tier 3 — 5 points each

Q11. A) Apple — Bob = cherry and Dan = date are given directly. That leaves apple and banana for Ann and Cara. Ann doesn't like banana, so Ann must like apple — and that leaves banana for Cara, which is consistent with "Cara doesn't like apple."

Q12. A) 1/16 — Each quadrant is 1/4 of the big square. The shaded triangle is 1 of 4 equal triangles inside that quadrant, so it's 1/4 of 1/4 = 1/16 of the whole big square.

Q13. C) 9 — The 2-digit numbers whose digits add to 9 are: 18, 27, 36, 45, 54, 63, 72, 81, 90 — that's 9 numbers.

Q14. D) S — In a row of 4 squares that wraps around to form 4 sides of a cube, the 1st and 3rd squares end up opposite each other, and so do the 2nd and 4th. Here the row is Q, X, R, S — so X (2nd) is opposite S (4th). (P and T, attached above and below X, become the cube's top and bottom faces, and end up opposite each other instead.)

Q15. B) Day 8 — Track the snail's height at the *top* of each day's climb (before any night-time slide): Day 1 → 3 m (slides back to 1 m), Day 2 → 4 m (back to 2 m), Day 3 → 5 m (back to 3 m), Day 4 → 6 m (back to 4 m), Day 5 → 7 m (back to 5 m), Day 6 → 8 m (back to 6 m), Day 7 → 9 m (back to 7 m), Day 8 → 7 + 3 = 10 m — exactly the top, so it climbs out and stops. This is a classic puzzle precisely because the "naive" answer (10 days, from 10 ÷ 1 m net gain per day) is wrong — the snail is already out before the final slide-back would even happen.

Scoring: This stretch set is intentionally harder and more varied than anything in the core plan — there's no "redo this before moving on" rule here. If you're getting most of Tier 1 and Tier 2 right, that's a strong result; Tier 3 questions are meant to make you stop and think, even if you don't get them all on the first try.
